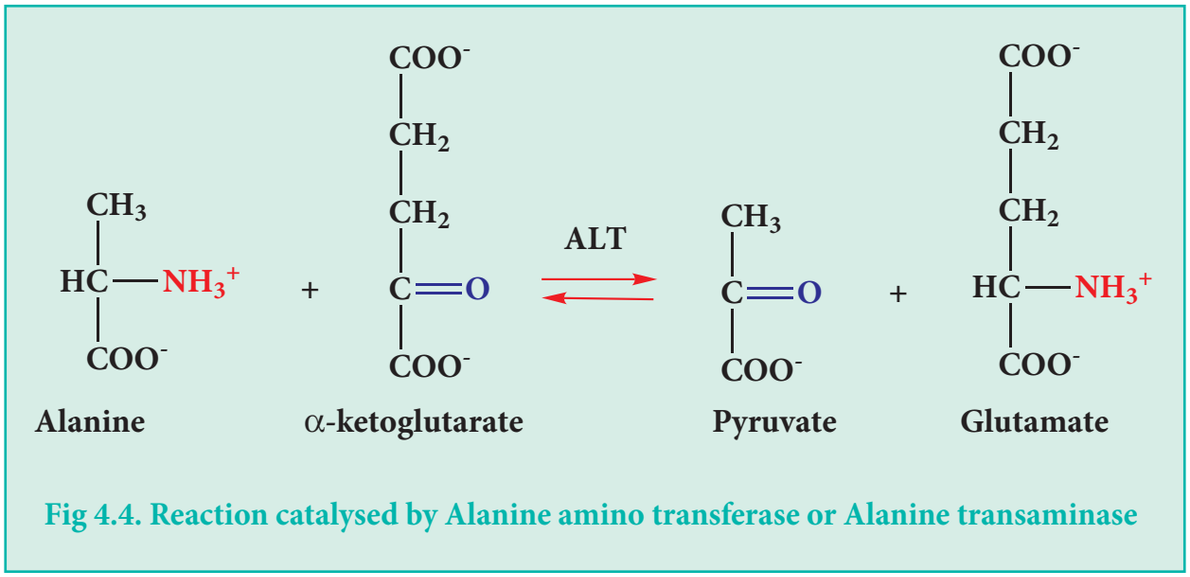


Fig 4.4. Reaction catalysed by Alanine amino transferase or Alanine transaminase

Transamination involves the coenzyme pyridoxal phosphate for the shuttling of amino groups. During this process, pyridoxamine phosphate is formed as an intermediate. The amino group of the amino acid reacts with pyridoxal phosphate bound to the enzyme to form a Schiff's base. As the alpha ketoglutarate is involved in all the transamination reactions, glutamate acts as a sink for the amino groups for most of the amino acids (Figure 4.5).

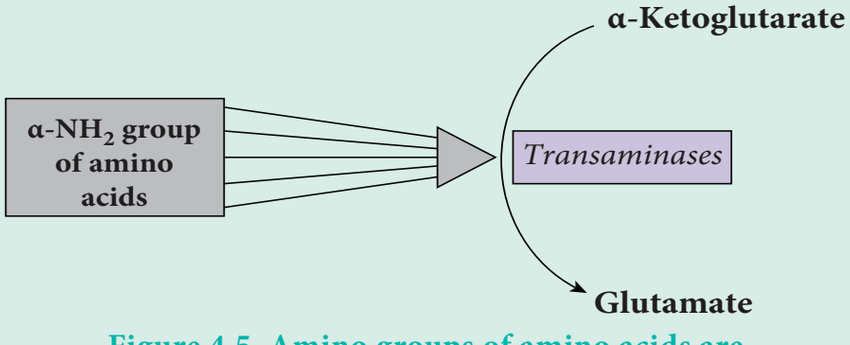


Figure 4.5. Amino groups of amino acids are accepted by alpha ketoglutarate

Features of Transamination

- Transamination reactions are reversible reactions.
- All amino acids except lysine, threonine, proline and hydroxyl proline undergo transamination.
- Serum transaminases are significant markers for diagnosis of liver and heart diseases.
- Many non-essential amino acids are synthesized using transamination reactions.

4.1.3 Deamination

Deamination is the process of removal of amino groups from amino acids as ammonia. Deamination may be or may not be coupled with oxidation. Depending upon this, deamination is classified as oxidative deamination and non-oxidative deamination.

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